ADITYA UNIVERSITY

PROBABILITY& STATISTICS

Assignment-I

Unit-I

1. A box contains 6 red balls and 4 green balls. Two balls are drawn at random, one after the other, without replacement. What is the probability that the second ball drawn is green, given that the first ball drawn was red?
2. A factory produces 3 types of machines: Type A, Type B, and Type C. The probability of producing each type of machine is as follows P(A) = 0.5, P(B) = 0.3, P(C) = 0.2 The probability of a machine being defective given its type is P(D | A) = 0.02, P(D | B) = 0.05, P(D | C) = 0.01. A machine is selected at random, and it is found to be defective. What is the probability that the defective machine is of Type A?
3. A bag A contains 2 white and 3 red balls and bag B contains 4 white and 5 red balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that the red ball drawn is from bag B.
4. A business man goes to hotels X, Y, Z, 20%, 50% and 30% of the time respectively. It is shown that 5%, 4% and 8% of the rooms in X, Y, Z hotels have faulty plumbing. What is the probability that business man's room having faulty plumbing is assigned to hotel X?
5. Suppose 5 men out of 100 and 25 women out of 10,000 are colour blind. A colour blind person is chosen at random. What is the probability that the person being male (Assume male and female to be in equal numbers).
6. A random variable X has the following probability function

X	0	1	2	3	4	5	6	7
P(x)	0	K	2k	2k	3k	K^2	$2k^2$	$7k^2+k$

Determine i) K ii) $P(X < 6)$, $P(X \geq 6)$ and $P(0 < X < 5)$
iii) if $P(X \leq k) > 1/2$, find the minimum value of k
iv) Standard Deviation of X and V) cumulative distribution of X.

7. Two dice are thrown. Let X assign to each point (a, b) in S the maximum of its numbers i.e., $X(a, b) = max.(a, b)$. Find the probability distribution. X is a random variable with $X(S) = \{1, 2,3, 4, 5, 6\}$. Also find the mean and variance of the distribution.
8. For the following probability distribution

X	-3	6	9
P(X)	1/6	1/2	1/3

Find i) E(X) ii) $E(X^2)$ iii) $E(2X+1)^2$

9. The probability density function of a continuous random variable is given by
$f(x) = \begin{cases} k(1-x^2) & 0 < x < 1 \\ 0 & otherwise \end{cases}$
Find i) k ii) probability between 0.1 and 0.2 iii) standard deviation

10. A continuous random variable X has a pdf $f(x) = kx^2e^{-x},\ x \geq 0$. Find k, mean and variance.
11. A continuums random variable X has the following cumulative distribution function
$F(x) = \begin{cases} 0, & \text{if } x \leq 1, \\ k(x-1)^3, & \text{if } 1 \leq x \leq 3, \\ 1, & \text{if } x > 3 \end{cases}$ Find i) k ii) p.d.f of X iii) mean.
12. Let the random variable X have the p.d.f $f(x) = \begin{cases} \frac{1}{2}e^{-\frac{x}{2}}, & x > 0 \\ 0, & otherwise \end{cases}$. Find the moment generating function, mean & Variance of X.

Unit-II

1. Determine the binomial distribution for which the mean is 4 and variance is 3.
2. Out of 800 families with 5 children each how many would you expect to have (a) 3 boys (b) 5 girls (c) either 2 or 3 boys (d) at least one boy (Assume equal probabilities for boys and girls.
3. The average number of homes sold by the Realty Company is 2 homes per day. Determine the probability that will be sold tomorrow i) exactly 3 homes, ii) in between 0 and 2, both inclusive?
4. The average number of phone calls per minute coming into a switch board between 2 p.m. and 4 p.m. is 2.5. Determine the probability that during one particular minute there will be i) 4 or fewer ii) more than 6 calls.
5. If the probability that an individual suffers a bad reaction due to a certain injection is 0.001. Determine the probability that out of 2000 individuals i) Exactly 3 and ii) More than 2 individuals will suffer a bad reaction.
6. If the masses of 300 students are normally distributed with mean 68 kgs and standard deviation 3 kgs, how many students have masses i) greater than 72 kg ii) less than or equal to 64 kg.
7. A company pays its employees an average wage of ₹350 per hour with a standard deviation of ₹30. If the wages are approximately normally distributed, compute the proportion of the workers getting i) wages between ₹325 and ₹360 per hour, ii) wage is more than ₹335, iii) at least wage is ₹380.
8. If X is poisson variate such that $3P(x = 4) = \frac{1}{2}p(x = 2) + P(x = 0)$. Find i) the mean of X (ii) $p(x \leq 2)$.
9. If X is a normal variate with mean 30 and standard deviation 5. Find i)$p(26 \leq X \leq 40)$, ii) $p(X \geq 45)$.
10. The length of time a person speaks over phone follows exponential distribution with mean 6. What is the probability that the person will talk for (i) more than 8 minutes (ii) between 4 and 8 minutes?